Cybersecurity Assessment 01

Task A | Secure a student account 25 marks

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A1. Five risks and CIA impacts (10 marks)
A2. Controls and limitations (10 marks)

C = confidentiality (preventing unauthorised disclosure); I = integrity (preventing unauthorised changes); A = availability (maintaining access).

Risk	Asset	CIA impact	Control	Limitation
1. Short, reused password: guessing or a breach can compromise both accounts.	Email, portal account and personal records.	C: messages exposed. I: submissions altered. A: account lockout.	Use a password manager to create long, random, unique passwords; one leaked password then cannot unlock both accounts. [1]	The manager needs a strong master password and secure recovery.
2. Approving unexpected sign-in prompts can admit an attacker.	Account sessions and student identity.	C: private records read. I: attacker impersonates Mira.	Reject and report unsolicited MFA prompts; approve only a login she initiated. This prevents accidental approval of attacker access. [2]	Relies on attention; previously approved sessions may remain active.
3. Delayed updates leave known vulnerabilities exposed.	Laptop, applications and stored data.	C: malware steals files. I: files modified. A: ransomware blocks access.	Enable automatic OS and application updates, completing required restarts to apply security fixes. [3]	Updates cannot eliminate unknown vulnerabilities and may briefly interrupt work.
4. One project copy creates a single point of failure.	Project files and submission deadline.	A: theft or disk failure loses access. I: corruption destroys usable work.	Keep automatic versioned backups plus a disconnected copy; test restoration so clean work can be recovered. [4]	Backups can be outdated; untested restores may fail.
5. A friend uses Mira s authenticated session.	Private files, messages and portal submissions.	C: friend views private content. I: changes appear to be Mira s actions.	Provide a separate standard user account and lock Mira s session before sharing; this separates access and activity.	Mira must consistently lock her session; deliberately shared files remain accessible.

Task A | Priorities and access control

A2. First three actions and justification

1. Stop approving unexpected prompts immediately. They could indicate an active login attempt. If Mira already approved one, use a trusted device to revoke unfamiliar sessions and contact portal support.

2. Replace both passwords, starting with email. Unique passwords stop reuse-based compromise. Email comes first because it commonly receives password-reset links for other accounts.

3. Back up the project and verify a restore. Its only copy is irreplaceable; this quickly reduces the consequences of theft, failure or corruption. Then promptly install updates and configure the friend's separate account before the next shared use.

A3. Authentication versus authorisation (5 marks)

Authentication verifies who is signing in: Mira supplies her portal credentials and an additional factor to establish her identity. [5]

Authorisation determines what that authenticated identity may do. Mira's student role should permit viewing her own grades and submitting her own coursework, but deny changing grades or accessing another student's private records. Permissions should be checked for every protected request. [6]

A friend using her existing session inherits Mira's access without proving their own identity, illustrating why session locking matters.

References and AI-use declaration

References

[1] CISA. *Cybersecurity Awareness Month 2022: Public Toolkit*. Accessed 26 October 2026. https://www.cisa.gov/sites/default/files/publications/CAM22_PublicToolkit_FINAL_OCC_CSD_DIR_508c.pdf

[2] CISA. *Turn On MFA*. Accessed 26 October 2026. https://www.cisa.gov/secure-our-world/turn-mfa

[3] FBI. *Cybersecurity Best Practices*. Accessed 26 October 2026. https://www.fbi.gov/investigate/cyber/cybersecurity-best-practices

[4] CISA. *#StopRansomware Guide*. Accessed 26 October 2026. https://www.cisa.gov/stopransomware/ransomware-guide

[5] NIST CSRC. *Glossary: Authentication*. Accessed 26 October 2026. https://csrc.nist.gov/glossary/term/authentication

[6] NIST CSRC. *Glossary: Authorization*. Accessed 26 October 2026. https://csrc.nist.gov/glossary/term/authorization